



Product Description

FNR

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1. General Information

floLIVE's Connectivity Management Platform (CMP) is a cloud-based solution that encompasses a fully virtualized 3G/4G/LTE and 5G core network as well as a complementary suite of Business Support Systems (BSS) that enable our customers to take full ownership over their IoT business, be it device manufacturing and distribution or complete IoT connectivity management.

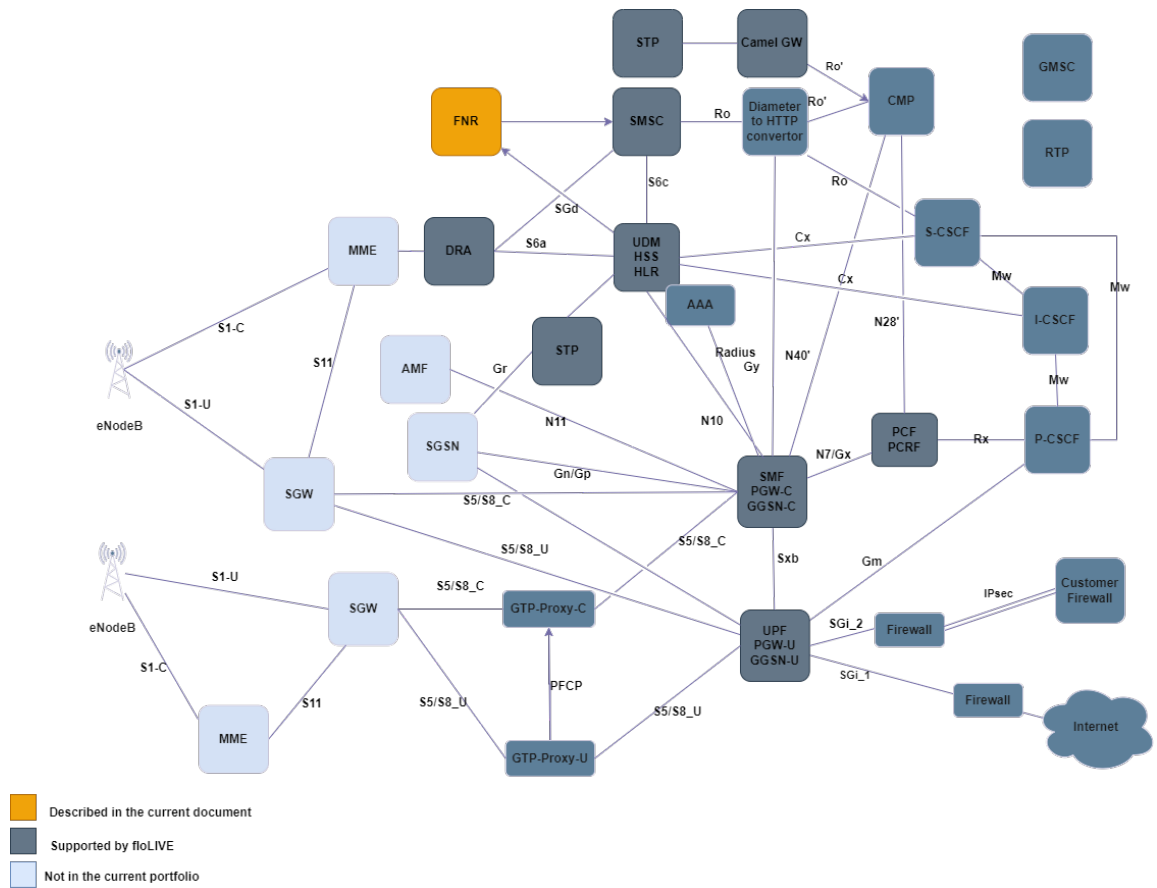
The platform provides the functionality necessary to support various workflows in managing the connectivity aspects of modern IoT and M2M device arrays.

floLIVE's CMP-as-a-Service solution comprises several subsystems and management applications.

The subsystems in use are the following:

- **floCONNECT Connectivity Management** - used for setting and managing accounts, users, SIMs, Self-care, Rules & Alerts, and more.
- **HLR/HSS** - Used for managing subscriber profiles, RAT Types, IMSI/MSISDN ranges and more.
- **GGSN/PGW(UPF & SMF)** - used for configuration and management of APNs, IP Pools, routing tables, etc.
- **SMSC** - used for configuration of SMS logic, forwarding rules and policies, external SMPP connections, and more.
- **floSIM Multi IMSI SIM Management** - used for configuration, management and control of SIM cards, business logic, PLMN definitions, OTA transactions, key management, and more.

Figure 1 - FNR



2. Overview

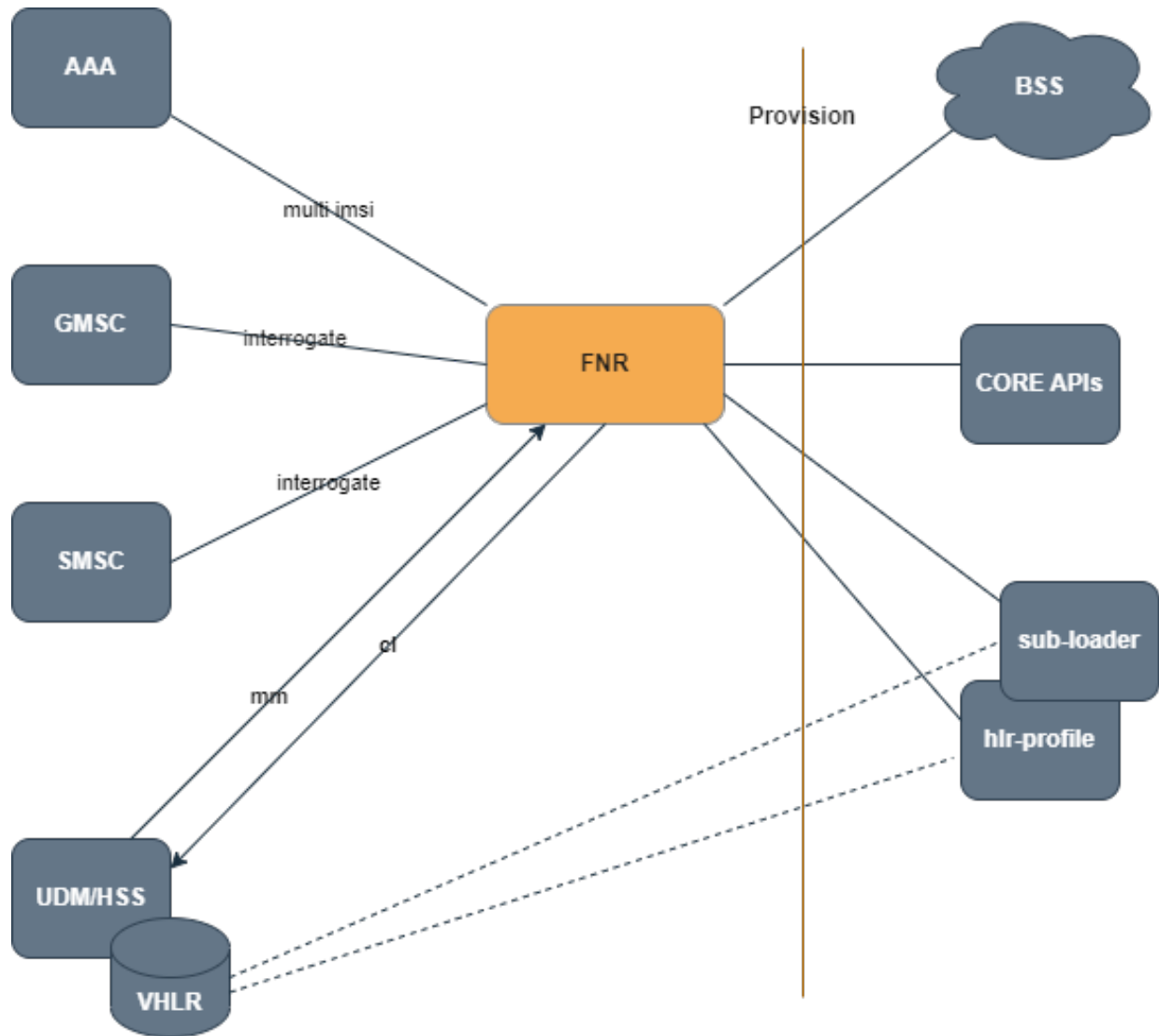
Flexible Number Register (FNR) is adding multi-identity feature to core network. It stores current IMSI of the customer and the nodes where it is registered - MME, AMF, VLR, SGSN. It also is used for provisioning of subscribers in FNR and VLR databases.

FNR is using REST API-s and Kafka for communications with the other components.

3. FNR Connectivity

FNR keeps track of a subscriber's active identity (IMSI) and the network node currently serving that subscriber - whether that is a 4G MME, 5G AMF, 3G/2G VLR, or SGSN. It also supports provisioning subscriber records into the register, so that other network functions can rely on it as a consistent source of subscriber location and identity information.

Figure 2 - Connectivity Diagram



FNR exposes REST APIs and streams events, allowing it to integrate with other systems such as HSS/UDM platforms, SMSC/GMSC nodes, and floLIVE's SIM management platform.

4. Integration and API

FNR is designed to integrate cleanly into an operator's existing core network and OSS/BSS systems through standard, well-documented interfaces.

Capability Area	Access Method	Description
Subscriber lookup	REST (GET/POST)	Resolve a subscriber's current IMSI, ICCID, and registration state from an MSISDN, IMSI, or ICCID.
Subscriber provisioning	REST (POST) / bulk file transfer (HTTP or UDP)	Add, update, or remove subscriber records, individually or in bulk.
Management commands	REST (POST)	Forward management commands (e.g., APN queries) to the subscriber's associated profile, for a single identity or an entire SIM.
Subscriber trace	REST (POST)	Activate or deactivate diagnostic tracing for a subscriber identity or ICCID, including redundant node-pair configurations.
Mobility events	Event streaming (Kafka)	Near real-time notifications of registration, attach, and detach events for dependent systems.

4.1. Data formats

- JSON — default format, supported for both submitting requests and reading responses.
- XML — supported for submitting requests.

4.2. Authentication

API calls that modify subscriber data or trigger management /trace actions require a bearer-token authorization header.

5. Key Capabilities

5.1. Mobility management

FNR tracks subscriber mobility across the network by:

- Receiving registration and location update events from HSS/UDM and updating subscriber records accordingly.
- Notifying floLIVE's SIM management platform of relevant mobility events.
- Serving lookup (interrogation) requests from SMSC, GMSC, and HSS/UDM.
- Triggering a cancel-location request to the previous network when a subscriber registers on a new one, ensuring only one network holds an active registration at a time.

5.2. Multi-identity support

FNR supports subscribers that may register under more than one IMSI on the same physical SIM (multi-IMSI), correctly tracking which identity is currently active.

5.3. Subscriber lookup (interrogation)

Given a subscriber identifier — MSISDN, IMSI, or ICCID — FNR returns the associated identifiers and current registration details, such as the corresponding IMSI, ICCID, or session control address.

5.4. Subscriber provisioning

FNR supports adding and removing subscriber records, either individually via the API or in bulk via file transfer. This keeps FNR's register synchronized as subscribers are onboarded or removed.

5.5. Subscriber trace

FNR can activate or deactivate diagnostic tracing for a subscriber. Tracing can be requested for a single identity or for an entire SIM (all IMSIs sharing an ICCID), and supports redundant pairs of destination nodes so that tracing remains available in high-availability deployments.

6. Example API Usage

The examples below illustrate the request format for common FNR API calls. Hostnames, ports, and tokens are placeholders.

6.1. Management command

Request a subscriber's APN configuration (JSON):

```
curl -d @request.json --request POST \  
-H "Content-Type: application/json" \  
-H "Authorization: Bearer <ACCESS_TOKEN>" \  
https://<fnr-host>/mng  
request.json:  
{  
  "request": {  
    "command": "getSubscriberAPNs",  
    "imsi": "<IMSI>"  
  }  
}
```

Equivalent request in XML:

```
<request origin="abc" id="1234">  
<command>getSubscriberEpsAPNs</command>  
<data_record>  
<imsi><IMSI></imsi>  
</data_record>  
</request>
```

6.2. Subscriber trace activation

Activate tracing for a subscriber (JSON):

```
{  
  "imsi": "<IMSI>",  
  "iccid": "<ICCID>",  
  "ip": "<TARGET_IP>",  
  "action": "TRACE_ON"  
}
```

Equivalent request in XML:

```
<request>  
<imsi><IMSI></imsi>
```

```
<iccid><ICCID></iccid>  
<ip><TARGET_IP></ip>  
<action>TRACE_ON</action>  
</request>
```

To deactivate tracing, submit the same request with "action": "TRACE_OFF".